

Parameter-Efficient Adaptation of SAM 2 for Sentinel-1 SAR Flood Mapping in Bangladesh

A Literature Review and Project Proposal

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Computer Vision Project: Literature Review Presentation

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The Problem: Bangladesh Floods Every Year

The reality on the ground:

- 20 to 25% of Bangladesh inundates in typical monsoons; 30 to 60% in severe years
- 2022 Sylhet flood: 7.2 million affected, ~480,000 displaced to shelters
- Ganges-Brahmaputra-Meghna delta: world's largest delta system

What's needed:

- Rapid, accurate flood-extent maps
- Available within hours, not days
- Reliable in any weather condition

Operational stakes

Better flood maps directly translate into:

- Faster boat dispatch to villages
- Smarter relief truck routing
- Faster damage assessment
- Lives saved

Why Satellites? And Why Radar Specifically?

Optical Satellites (Sentinel-2)

- RGB-like multispectral imagery
- Easy to interpret visually
- **Useless during monsoon**
- Clouds block visible/IR light
- Cloud cover > 80% for weeks

Radar Satellites (Sentinel-1)

- Synthetic Aperture Radar (SAR)
- **Sees through clouds, day/night**
- Free, global, 6-day revisit
- 10m spatial resolution
- Dual-polarization (VV + VH)

Bottom line: During the South Asian monsoon, Sentinel-1 SAR is *the only* reliable imaging modality for flood mapping.

But SAR is Hard for Computer Vision

What makes SAR different:

- Speckle noise (multiplicative, not additive)
- Only 1 or 2 channels (not 3-channel RGB)
- Logarithmic dB scale, wide dynamic range
- Water = dark (specular reflection)
- Vegetation, buildings = bright (backscatter)

Why this breaks foundation models:

- Pretrained on natural RGB photos
- Never seen radar imagery
- Statistics are alien
- Zero-shot performance: poor

The opportunity

Adapt a powerful foundation model (Segment Anything Model 2) to this new modality **efficiently**, without millions of dollars of compute.

Our Four Research Questions

- RQ1 Adaptation method:** Which parameter-efficient fine-tuning (PEFT) approach works best for SAM 2 on Sentinel-1 SAR?
- RQ2 Polarimetric prompts:** How should we compose the dual-polarization channels (VV, VH) into the 3-channel input SAM expects?
- RQ3 Confidence:** Can we add calibrated uncertainty estimates to flood maps for operational decision support?
- RQ4 Generalization:** Does a globally-trained model work on the Ganges-Brahmaputra delta specifically?

Citation

Kirillov, A., Mintun, E., Ravi, N., *et al.* (2023). **Segment Anything**. *ICCV 2023* (Best Paper Honorable Mention). pp. 4015 to 4026.

[arXiv:2304.02643](https://arxiv.org/abs/2304.02643) ~17,000 citations

What they did:

- Built a foundation model for segmentation
- 3 components: image encoder (ViT), prompt encoder, mask decoder
- Created SA-1B: 1 billion masks, 11M images
- Released as open weights (Apache 2.0)

Key innovations:

- Promptable segmentation as a task
- Data engine with model-in-the-loop
- Strong zero-shot generalization
- Single model, many downstream tasks

Relevance to our project

This is our starting point. We adapt SAM (and its successor) to SAR flood mapping. Without SAM, no foundation-model-based approach exists for our task.

Citation

Ravi, N., Gabeur, V., Hu, Y.-T., *et al.* (2025). **SAM 2: Segment Anything in Images and Videos**. *ICLR 2025 (Oral)*.

[arXiv:2408.00714](https://arxiv.org/abs/2408.00714) ~2,700+ citations

Three big upgrades over SAM:

- **Hiera encoder**: multi-scale features
- **Memory module**: streaming attention across frames
- **6× faster, more accurate**: even on still images

Model variants (open weights):

- Hiera-Tiny: 39M parameters
- Hiera-Small: 46M
- **Hiera-Base+: 81M (our choice)**
- Hiera-Large: 224M

Relevance to our project

This is our chosen backbone. The memory module is a hidden gem: it was designed for video, but it perfectly handles **pre-flood vs. post-flood image pairs**, exactly the bi-temporal structure of SAR flood detection.

Citation

Zhong, Z., Tang, Z., He, T., Fang, H., Yuan, C. (2024).

Convolution Meets LoRA: Parameter Efficient Finetuning for Segment Anything Model.
ICLR 2024. [arXiv:2401.17868](https://arxiv.org/abs/2401.17868)

The core idea:

- Standard LoRA = low-rank linear updates
- Add ultra-light *convolutional* layers
- Inject **locality bias** into ViT encoder
- Trains <5% of model parameters

Why convolutions matter:

- Plain ViT lacks spatial locality
- Speckle in SAR is *very* local
- Water boundaries are local
- Convolutions match the inductive bias

Relevance to our project

This is our methodology blueprint. Conv-LoRA was tested on *remote sensing* and *medical imaging*, both domains where SAM's plain ViT struggles. SAR sits squarely in this category.

Conference Papers: At-a-Glance Comparison

Paper	Venue (Tier)	Year	Role in Our Project
SAM [1]	ICCV (A*)	2023	Foundation model we adapt
SAM 2 [2]	ICLR Oral (A*)	2025	Chosen backbone plus memory module for bi-temporal SAR
Conv-LoRA [3]	ICLR (A*)	2024	Adaptation method blueprint

Narrative arc

Foundation (SAM) → **Upgrade** (SAM 2) → **Method** (Conv-LoRA).

Three papers, three roles, one coherent technical foundation.

Citation

Ma, J., He, Y., Li, F., Han, L., You, C., Wang, B. (2024).

Segment Anything in Medical Images.

Nature Communications, 15, 654. IF: 15.7 (Q1, Multidisciplinary Sciences)

What they did:

- Adapted SAM to medical imaging
- Curated 1.57M mask annotations
- Spans CT, MRI, ultrasound, X-ray, microscopy
- Bounding-box-prompted segmentation
- Released model and dataset publicly

Why this is the gold-standard template:

- First major SAM adaptation to a non-RGB modality
- Showed: pretrained features *do* transfer
- Validated by $\sim 2,800$ citations
- Published in a top-IF journal

Relevance to our project

This is the writeup template our committee will recognize. The story of “adapt SAM to a domain it was never designed for” is now well-established. We replicate the framing for SAR using PEFT instead of full fine-tuning.

Citation

Saleh, T., Weng, X., Holail, S., Hao, C., Xia, G.-S. (2024).

DAM-Net: Flood detection from SAR imagery using differential attention metric-based vision transformers.

ISPRS Journal of Photogrammetry and Remote Sensing, 212, 440 to 453. JCR 2024 IF: 12.2 (Q1)

Architecture:

- Siamese Vision Transformer
- Pre-flood + post-flood SAR pairs
- Temporal differential fusion module
- Class token records semantic change

Performance on S1GFloods:

- F1: **96.5%**
- IoU: **93.2%**
- Released the S1GFloods dataset
- Current SOTA on bi-temporal SAR

Relevance to our project

This is the SOTA non-foundation-model baseline we must beat (or come close to). DAM-Net validates the bi-temporal Siamese architecture, which we replicate using SAM 2's memory module instead of an explicit Siamese network.

Citation

Pu, X., Jia, H., Zheng, L., Wang, F., Xu, F. (2025).

ClassWise-SAM-Adapter: Parameter-Efficient Fine-Tuning Adapts Segment Anything to SAR Domain for Semantic Segmentation.

IEEE J. Selected Topics in Applied Earth Observations and Remote Sensing, 18, 4791 to 4806. (Q1)

What they did:

- Froze most of SAM's parameters
- Added lightweight adapters to encoder
- Built a class-aware mask decoder
- Frequency-domain task-specific input
- **Target: SAR landcover classification**

Why it matters and how we differ:

- **Same:** SAM + adapter + SAR
- **Different:** flood mapping (not landcover)
- **Different:** SAM 2 (not SAM v1)
- **Different:** confidence-aware
- **Different:** Bangladesh test set

Relevance to our project

This is our nearest neighbor in the literature. Our four differentiators above are precisely the **novelty wedge** that distinguishes our thesis from CWSAM.

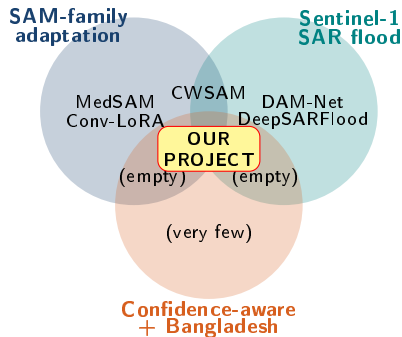
Journal Papers: At-a-Glance Comparison

Paper	Journal (IF, Quartile)	Year	Role in Our Project
MedSAM [4]	Nat. Comm. (15.7, Q1)	2024	Cross-domain adaptation template
DAM-Net [5]	ISPRS-J P&RS (12.2, Q1)	2024	Non-FM SOTA baseline to beat
CWSAM [6]	IEEE J-STARS (Q1)	2025	Closest prior work; novelty wedge

Narrative arc

How to write up (MedSAM) → **What to beat** (DAM-Net) → **What we extend** (CWSAM).
Together with the conference papers, we have a complete view of the field.

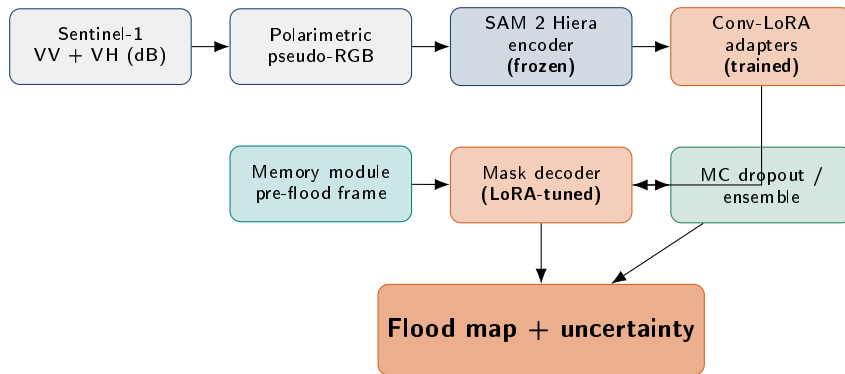
The Research Gap, Visualized



The triple intersection is empty, and that's our contribution

No published peer-reviewed work combines all three: **SAM-family adaptation** $\cap$ **Sentinel-1 SAR flood mapping** $\cap$ **confidence-aware regional benchmark**.

End-to-End Pipeline



Key design choices

Frozen pretrained encoder + trainable Conv-LoRA adapters + SAM 2 memory module for bi-temporal context + Monte-Carlo dropout for calibrated uncertainty.

Methodology: Four Experimental Pillars

Pillar 1 PEFT comparison: 4 methods (LoRA, DoRA, Conv-LoRA, AdaptFormer) $\times$ 2 backbones (SAM ViT-B, SAM 2 Hiera-B+) $\times$ 3 seeds = **24 runs**

Pillar 2 Polarimetric prompt ablation: 3 channel compositions

- (VV, VH, VV/VH ratio): industry convention
- (VV, VH, VV-VH difference): physically interpretable
- (VV, VV, VV): single-pol control

Pillar 3 Confidence-aware mechanism: Monte Carlo dropout (20 passes) or 5-seed deep ensemble; calibration via expected calibration error and reliability diagrams

Pillar 4 Out-of-distribution generalization: Sen1Floods11 hand-labeled test $\rightarrow$ **Bolivia held-out split** (the only truly held-out country in Sen1Floods11)

Datasets We Will Use

Dataset	Use	Size	Notes
Sen1Floods11 hand	Train, val, in-dist test	700 MB	446 hand-labeled chips across 10 countries; canonical SAR flood benchmark
Sen1Floods11 Bolivia	OOD held-out test	subset of above	15 chips; only truly held-out country, used by virtually every paper in the area

Released alongside this thesis

A **Bangladesh-2022-Sylhet dataset construction pipeline** (UNOSAT + Copernicus GFM + optional Charter 762) and the **Pakistan-2022 TU Wien** flood-extent release ship as standalone artifacts. Empirical evaluation on both is deferred to subsequent work.

12-Week Project Timeline

Weeks	Activities
1 to 2	Literature consolidation, environment setup, sanity-check inference
3	Sen1Floods11 download, PyTorch Dataset class, augmentation pipeline
4	Bangladesh-2022-Sylhet test set construction
5	U-Net baseline reproduction, zero-shot SAM baseline, locked protocol
6 to 7	Core PEFT comparison: 4 methods, 2 backbones, 3 seeds
8	Polarimetric prompt-engineering ablation
9	Confidence-aware mechanism + calibration analysis
10	Vast.ai final sweep including SAM ViT-L stretch (budget <\$25)
11 to 12	Analysis, figures, thesis writeup, presentation prep, repo freeze

Hardware: realistic and on-hand

RTX 3060 12GB (local) for 95% of training, plus rented RTX 4090 24GB (Vast.ai, <\$25 total) for the final sweep including the largest backbone.

Expected Contributions

- C1** Systematic empirical comparison of **4 PEFT methods** on **2 SAM-family backbones** for Sentinel-1 SAR flood mapping: the first such comparison at peer-reviewed quality.
- C2** A polarimetric prompt-engineering protocol with a defensible recommendation for composing dual-polarization SAR into 3-channel pseudo-RGB.
- C3** The **first application of a confidence-aware mechanism to SAR flood mapping with SAM 2**, with calibration analysis stratified by terrain type.
- C4** A standalone **Bangladesh-2022-Sylhet dataset construction pipeline** released alongside the thesis, with empirical evaluation deferred to subsequent work.
- C5** Open-source release of all code, configurations, and trained models for full reproducibility.

Risks and Mitigations

Risk	Mitigation
A PEFT method fails to converge	Drop from comparison; report on remaining methods; document the failure as a finding
Cloud-GPU budget exceeded	Reduce to one backbone and one PEFT method per ablation; mark missing rows as deferred to subsequent work
Competing work appears during the project	Pivot framing toward strongest distinguishing components: confidence mechanism, released construction pipeline
Foundation model loses to U-Net (PANGAEA effect)	Report honestly; argue that PEFT closes the compute-efficiency gap even if not the performance gap

Summary: What We Set Out to Do

In one sentence

We adapt SAM 2 to Sentinel-1 SAR for flood mapping using parameter-efficient fine-tuning, add a confidence-aware mechanism, evaluate on Sen1Floods11 + Bolivia + Pakistan-2022, and release a Bangladesh test set construction pipeline as a standalone artifact.

Why it matters:

- **Operational:** Bangladesh and the wider Indo-Gangetic plain need better monsoon flood maps; SAR is the only reliable modality
- **Scientific:** Closes a real gap at the intersection of foundation models and SAR remote sensing
- **Methodological:** Demonstrates PEFT-based foundation-model adaptation on consumer hardware
- **Reproducible:** Open-source pipeline lets future work extend evaluation to regional test sets

Six papers reviewed → One coherent project → Four defensible contributions.

References: The Six Reviewed Papers



A. Kirillov *et al.*, “Segment Anything,” *ICCV*, 2023, pp. 4015 to 4026. arXiv:2304.02643.



N. Ravi *et al.*, “SAM 2: Segment Anything in Images and Videos,” *ICLR (Oral)*, 2025. arXiv:2408.00714.



Z. Zhong *et al.*, “Convolution Meets LoRA: Parameter Efficient Finetuning for Segment Anything Model,” *ICLR*, 2024. arXiv:2401.17868.



J. Ma *et al.*, “Segment Anything in Medical Images,” *Nature Communications*, 15:654, 2024.



T. Saleh *et al.*, “DAM-Net: Flood detection from SAR imagery using differential attention metric-based vision transformers,” *ISPRS Journal of Photogrammetry and Remote Sensing*, 212:440 to 453, 2024.



X. Pu *et al.*, “ClassWise-SAM-Adapter: Parameter-Efficient Fine-Tuning Adapts Segment Anything to SAR Domain for Semantic Segmentation,” *IEEE J-STARS*, 18:4791 to 4806, 2025.

Thank You